

ANVAY VATS

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EDUCATION

University of Waterloo (Triple Major)

Expected April 2026

Bachelor of Computer Science, Combinatorics and Optimization, and Pure Mathematics

- Clubs: Wat.AI, Wat Street (Quantitative Finance), CS Club, Data Science Club, Pure Math Club
- CS Courses: DS & Algorithms, Operating Systems, Assemblers (RISC-V), Neural Networks, Applied Cryptography
- Math Courses (**all advanced**): Calculus I, II, & III; Linear Algebra I & II; Probability, Statistics, Groups, Optimization
- Scholarships: **\$25000 Global Math Scholarship** to study Computer Science at the University of Waterloo

EXPERIENCE

Founding AI Engineer | [Cupertino Labs \(Vercel AI Accelerator S25\)](#)

April 2025 – August 2025

CrewAI, LangGraph, MCP, Manus, tRPC, Jenkins, SonarQube, SageMaker, Airflow

San Francisco

- Shipped the analytics engine used on **10 million+ smartphones worldwide**, in contracts with AT&T and Airtel
- Replaced ETL on GCP clusters with a **CrewAI + Manus multi-agent workflow**, cutting KPI reporting latency by **90%**
- Built **DevSecOps Jenkins pipelines** integrating SonarQube (coverage, SCA, quality gates), Fortify, and JFrog Xray

Data Engineer | [DataCurve \(YC W24\)](#)

October 2024 – December 2024

Elixir, Ecto, GraphQL, TypeScript, NextJS, TablePlus, Postman, Clerk

San Francisco

- Managed a **7-figure contract** to supply clean training data for **Cohere's Chain-of-Thought LLM evaluation**
- Applied Ecto Repos, queries, and schemas with Elixir resolvers and schemas to ingest **1000+ user submissions**
- Added features for users to do real-time voice recordings to more accurately emulate human thought processes

Lead Developer (Gen AI) | [Arvo AI \(Panache VC-backed\)](#)

Aug 2024 – Dec 2024

LangGraph, React, Ollama, llama.cpp, MLFlow, Weaviate, Docker, Terraform

Montreal

- Led design + development of a freight logistics scheduler on a **6-figure contract** using LangGraph and Ollama
- Made **PoWs, DoDs, GANTT, RACI, and RAID logs** and finished 150+ tickets on Linear in Agile Kanban sprints
- Delivered GNN-based freight optimizer and RAG pipeline with **90% precision, 95% recall, and 100 ms latency**

Quantitative Developer | [TIW Capital \(\\$200M+ AUM\)](#)

April 2024 – August 2024

Python, C++, Julia, Redis, Kafka, QuantLib, PostgreSQL, GoRoutines

Singapore

- Implemented **dynamic risk optimization** systems using **variable position sizing** and cross-sectional momentum strategies, reducing drawdowns by 32% while maintaining alpha
- Engineered high-frequency **adaptive trend following** algorithms with multi-asset **normalized signal processing**
- Developed **carry-enhanced, fast mean-reversion strategies** for break-out detection based in **deep diversification**

Graphics Developer | [MappedIn](#)

April 2023 – August 2023

TypeScript, Python (Jupyter), ThreeJS, WebGL2

Waterloo (Hybrid)

- Worked on SDK and ML teams developing 3D indoor maps used by **1000+ firms** like **Apple, Dubai Mall, and NFL**
- Applied rasterization, rendering, and projective geometry via SDK features and fixes using **ThreeJS**
- Researched and implemented **EdgeGAN with GNNs** to improve our **cGAN-based Map Digitizer for Microsoft**

AWARDS

Silver Medal in National Olympiad of Informatics (NOI)

February 2022

National University of Singapore (NUS), School of Computing

- Gold medalists were selected to be on [Singapore's International Olympiad in Informatics \(IOI\) team](#)

PROJECTS

[Dynamis Trading System \(\\$3.7M USD Monthly Volume Traded\)](#)

Rust, Python, Redis, Polars, JAX, AWS (ECS Fargate, EC2, Secrets Manager, TGW), NATS, CUDA

- Leading **full-stack development** in a team of 6 with the goal of **low-latency Solana trading strategies**
- Created the **fastest Twitter/X Solana token sniper in the world** with **25ms end-to-end purchase latency**
- Deployed **Langgraph agentic workflows** to transform plaintext trading strategies into **CFGs for parallel parsing**
- Architected data-intensive distributed system using **Jito gRPCs** and **NATS JetStreams** with microservices

TECHNICAL SKILLS

Languages: Python, Rust, C++20, SQL, Go, TypeScript, Julia, Elixir

AI/ML: PyTorch, JAX, HuggingFace Transformers & Diffusers, LangGraph, CrewAI, DSPy, MLFlow, Weaviate, Triton, CUDA

Quant/Data: Polars/pandas, DuckDB, Redis, Kafka, NATS, QuantLib, Risk & Statistical Modeling, Timeseries Forecasting

Interests: [Documenting how life shapes my beliefs](#), Startups, Philosophy, Poker, Mental math, Badminton, Stand-up